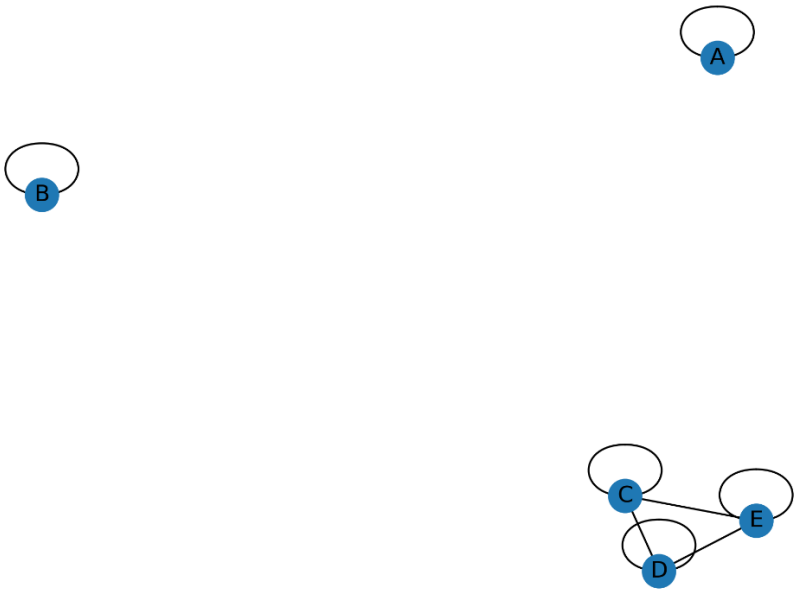
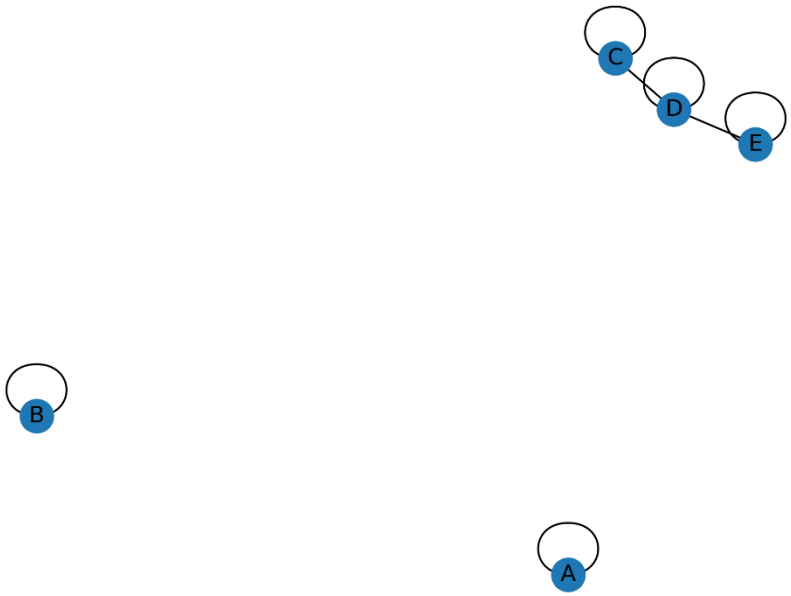
Task 1

Relation if euclidian distance is greater than 11

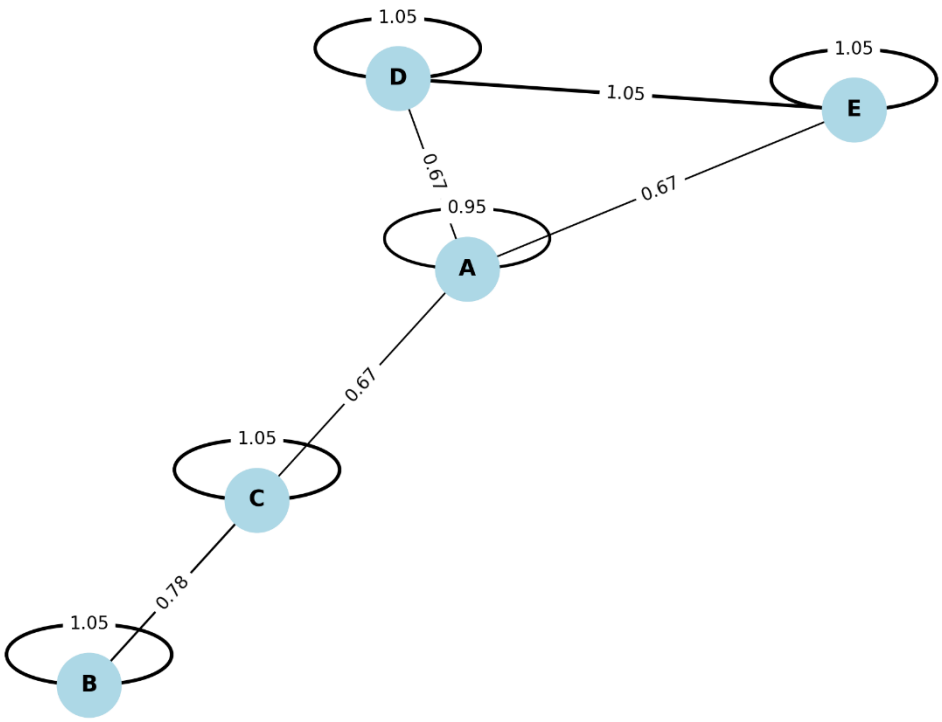


Relation if correlation coefficient is greater than 0.75 (in absolute value). Colored red if positive correlation

Relation if L1 form is smaller than 20



Relation if Mutual Information is greater than 0.65



Data	width	0,2375			width	0,215			
	x1	0,03	0,14	0,18	y1	0,09	0,15		
	x2	0,3	0,39	0,49	y2	0,48			
	x3	0,54	0,65		y3	0,53	0,65	0,71	
	x4	0,91	0,98		y4	0,74	0,8	0,88	0,95

Probability

p(x,y)	x1	x2	x3	x4	marginal p(y)
y1	0	0,1	0	0,1	0,2
y2	0	0	0,1	0	0,1
y3	0,1	0,1	0,1	0	0,3
y4	0,2	0,1	0	0,1	0,4
marginal p(x)	0,3	0,3	0,2	0,2	

p(x)p(y)	x1	x2	x3	x4	marginal p(y)
y1	0,06	0,06	0,04	0,04	0,2
y2	0,03	0,03	0,02	0,02	0,1
y3	0,09	0,09	0,06	0,06	0,3
y4	0,12	0,12	0,08	0,08	0,4
marginal p(x)	0,3	0,3	0,2	0,2	

Independence NO, because p(x,y) != p(x)p(y)

Conditional distribution x|y=3	0,33	0,33	0,33	0,00
Conditional distribution y|x=4	0,5	0	0	0,5

Entropy **3,122**

-1*p(x)*LOG(p(x);2)	x1	x2	x3	x4	marginal y	
y1	0,000	0,332	0,000	0,332	0,464	
y2	0,000	0,000	0,332	0,000	0,332	
y3	0,332	0,332	0,332	0,000	0,521	
y4	0,464	0,332	0,000	0,332	0,529	
marginal x	0,521	0,521	0,464	0,464		**1,971**
					1,846	

Conditional Entropies

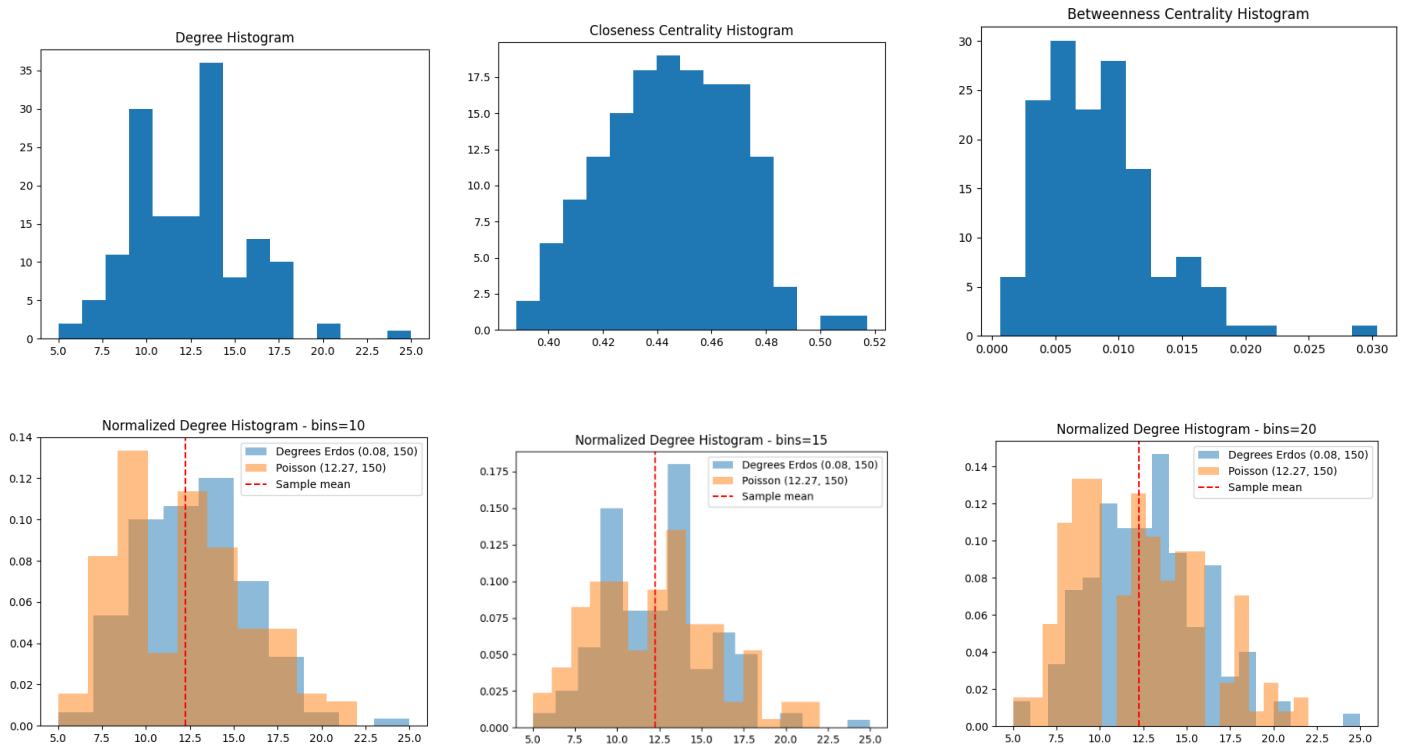
H(X|Y) = X(X,Y)-X(Y)	**1,275**
H(Y|X) = X(X,Y)-X(X)	**1,151**

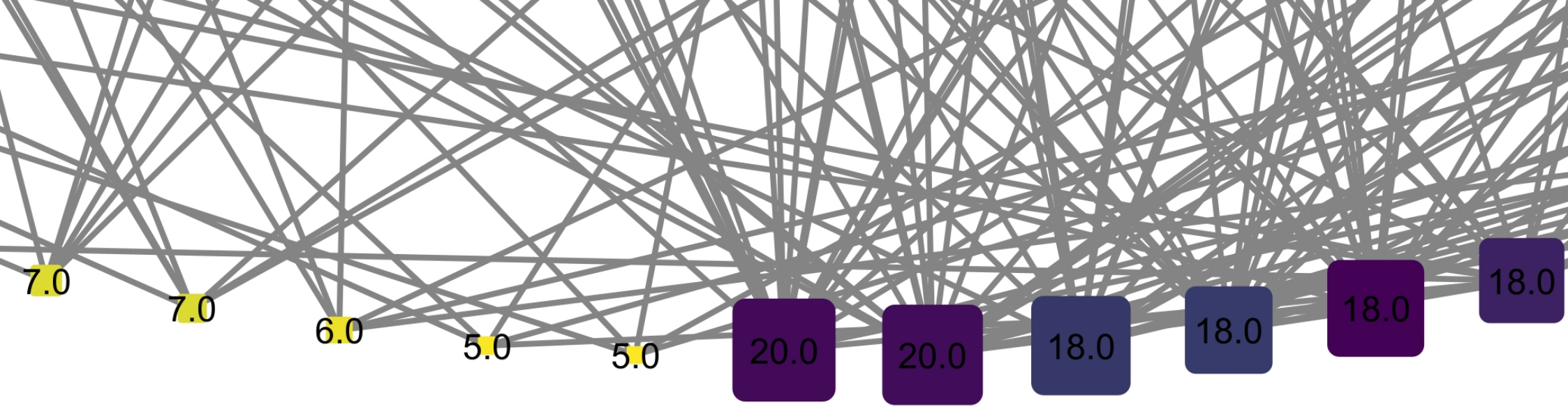
Mutual information

H(X)+H(Y)-H(X,Y)	**0,6955**	0,6955	x1	x2	x3	x4
Definition	**0,6955**	y1	0,0000	0,0737	0,0000	0,1322
		y2	0,0000	0,0000	0,2322	0,0000
		y3	0,0152	0,0152	0,0737	0,0000
		y4	0,1474	-0,0263	0,0000	0,0322

Task 3

1. Generate Erdos-Renyi G(n=150,p=0.08).
2. Analyze:
   - number and size of connected components
   - histograms for: degree value, closeness, and betweenness centralities
   - normalize degree histogram and plot Poisson PMS over it (scipy.stats.poisson)
   - compare for different bin numbers. Q Is the bin size important?
3. Export the network as GraphML file and attributes (centralities, degree) as csv file
4. Check what happens when different 'p' (no need to hand in)




7.0
7.0
6.0
5.0
5.0
20.0
20.0
18.0
18.0
18.0
18.0

$$\approx \boxed{0.9}$$

And by what we just showed, we get same result by applying the probability definition.

Q4 We can see from the plot :-

$$\boxed{\begin{aligned} f_X(x) &= 4x \quad \text{for} \quad 0 \le x \le 0.5 \\ f_X(x) &= 2 - 4(x - 0.5) \end{aligned}}$$

for $0.5 \le x \le 1$

$$E(X) = \int x f(x)\, dx =$$

$$\int_0^{0.5} x\,(4x)\,dx \quad +$$

$$\int_{0.5}^{1} x\,(4-4x)\,dx$$

$$= \left[\frac{4x^3}{3}\right]_0^{0.5}$$

$$+$$

$$\left[\frac{4x^2}{2} - \frac{4x^3}{3}\right]_{0.5}^{1}$$

$$= \frac{4\,(0.5)^3}{3}$$

$$+ \quad 2 \quad - \frac{4}{3} \quad - \frac{1}{2} \quad + \frac{4(0.5)^3}{3}$$

$$= \frac{12}{6} - \frac{8}{6} - \frac{3}{6} + \frac{2}{6}$$

$$= \boxed{\frac{1}{2}}$$

So, $\boxed{E(X) = \frac{1}{2}}$

$$E(X^2) = \int x^2 f(x)\, dx$$

$$= \int_0^{0.5} x^2 (4x)\, dx$$

$$+ \int_{0.5}^{1} x^2 (4 - 4x)\, dx$$

$$= \left[x^4 \right]_0^{0.5}$$

$$+ \left[\frac{4x^3}{3} - x^4 \right]_{0.5}^{1}$$

$$= (0.5)^4 + \frac{4}{3} - 1$$

$$- \frac{0.5}{3} + (0.5)^4$$

$$= \frac{1}{8} + \frac{4}{3} - 1 - \frac{1}{6}$$

$$= \frac{3}{24} + \frac{32}{24} - \frac{24}{24} - \frac{4}{24}$$

$$= \boxed{\frac{7}{24}}$$

So, $\boxed{E(X^2) = \frac{7}{24}}$

$$\boxed{Var(X)} = E(X^2) - [E(X)]^2$$

$$= \frac{7}{24} - \left(\frac{1}{2}\right)^2$$

$$= \boxed{\frac{1}{24}}$$